

Baoli Hao

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 [Baoli's Google Scholar](#)

EDUCATION

Illinois Institute of Technology

Ph.D. in Applied Mathematics

Aug. 2022 – Expected Apr. 2027

Chicago, IL

- **Advisors:** Prof. Chun Liu & Prof. Ming Zhong | **GPA:** 3.93/4.0

City University of Hong Kong

BSc in Computing Mathematics (First Class Honours)

Aug. 2017 – Apr. 2021

Hong Kong

- **GPA:** 3.7/4.3 | **Minor:** Computer Science

RESEARCH INTERESTS

My research focuses on **inverse problems** and **scientific machine learning (SciML)** for complex dynamical systems, with an emphasis on learning hidden mechanisms from incomplete observations by exploiting mathematical and physical structure. My work also connects these problems with **multiscale modeling**, **nonlinear PDEs**, and **interacting particle systems**, with the goal of developing stable, interpretable, and physically meaningful methods for data-driven inference and scientific computation.

PUBLICATIONS & PREPRINTS

1. **Baoli Hao**, Kamrun Mily, Annalisa Quaini, and Ming Zhong. A finite element framework for simulating residential burglary in realistic urban geometries. *Mathematical Models and Methods in Applied Sciences*, 36(5), 1019–1050, 2026.
2. Kin Shing Chan, **Baoli Hao**, Kei Fong Lam, and Björn Stinner. On a phase field model for binary mixtures of micropolar fluids with non-matched densities and moving contact lines. *Interfaces and Free Boundaries*, published online first, 2026.
3. **Baoli Hao**, Chun Liu, Ulisses Braga-Neto, Lifan Wang, and Ming Zhong. Stability in training PINNs for stiff PDEs: Why initial conditions matter. *Foundations of Data Science*, Early Access, 2026.
4. **Baoli Hao**, Ming Zhong, and Kevin O’Keeffe. Attractive and repulsive interactions in the one-dimensional swarmalator model. *Physical Review E*, 108(6), 064214, 2023.
5. **Baoli Hao**, Kamrun Mily, Annalisa Quaini, and Ming Zhong. Crime hotspot dynamics in residential burglary models with police response. arXiv:2605.17709, 2026.
6. **Baoli Hao**, Fanze Kong, Kei Fong Lam, and Chun Liu. Global finite-energy weak solutions and sharp entropy decay for a Poisson–Nernst–Planck system with interspecies drag and steric effects. arXiv:2607.21742, 2026.
7. **Baoli Hao**, Chenxi Hu, Ming Zhong, and Ren Wang. Event-structured physics-informed neural networks for differentiable critical clearing boundaries. arXiv:2607.27681, 2026.

MANUSCRIPTS

1. **Baoli Hao**, Mauro Maggioni, and Ming Zhong. One shot of learning of collective dynamics from equilibrium in interacting particle systems. Manuscript in preparation, 2026.
2. **Baoli Hao**, Mauro Maggioni, and Ming Zhong. Learning interaction topology and kernels in interacting particle systems. Manuscript in preparation, 2026.
3. **Baoli Hao**, Mauro Maggioni, and Ming Zhong. Data-driven discovery of gravitational structure from solar-system ephemerides. Manuscript in preparation, 2026.
4. **Baoli Hao**, Annalisa Quaini, and Ming Zhong. Discovery of hidden crime dynamics. Manuscript in preparation, 2026.

5. **Baoli Hao**, Xiaofan Li, Chun Liu, Yiwei Wang, and Ming Zhong. Data-driven coarse-graining of nonlocal steric interactions in ionic transport. Manuscript in preparation, 2026.

TALKS & PRESENTATIONS

SIAM Annual Meeting

Title: *Modeling, Simulation and Learning for Urban Crime Dynamics*

Jul. 2026
Cleveland, OH

SIAM Texas-Louisiana Sectional Meeting

Title: *Stability in training PINNs for stiff PDEs: Why initial conditions matter*

Sept. 2025
Austin, TX

Central AMS Sectional Meeting

Title: *Modeling and analysis of Poisson-Nernst-Planck systems with steric effects and relative drags*

Apr. 2025

Lawrence, KS

INTERNSHIPS

HKU Business School, The University of Hong Kong

Data Scientist

Jul. 2021 – May 2022
Hong Kong

- Supervisor: Dr. Qingchen Wang
- Developed machine-learning and predictive models for risk stratification and survival analysis using large-scale electronic patient record data covering approximately 90% of Hong Kongs population.

SERVICE

Co-organizer, SIAM Great Lake Sectional Meeting, Purdue University

Sept. 2026

Coordinator, Multiscale Modeling and Computation Seminar, Illinois Tech

Aug. 2022 – Present

Sponsorship and Student Support Chair, ACM HyperText 2025 Conference, Illinois Tech

Oct. 2024 – Sept. 2025

Reviewer, ACM Transactions on Modeling and Computer Simulation (TOMACS)

Jan. 2026

Program Committee Member, AAAI Conference on Artificial Intelligence (AAAI 2027)

Aug. 2026

Reviewer, Studies in Applied Mathematics

Sept. 2026

TEACHING

Illinois Institute of Technology

MATH Graduate Qualifying Exam Analysis Prep Session (*Leader & Mentor*)

Fall 2025 – Present

MATH 500: Applied Analysis I (*Teaching Assistant*)

Fall 2025

MATH 578: Computational Mathematics II (*Grader*)

Spring 2025

MATH 151: Calculus I (*Recitation Instructor*)

Spring 2023

HONORS & AWARDS

Menger Lecture Poster Award, Illinois Institute of Technology

Apr. 2026

Full Tuition Scholarship, City University of Hong Kong

2017 – 2021

Deans List, City University of Hong Kong

2018 – 2021

SKILLS

Programming : Python, R, MATLAB, SQL

Frameworks & Libraries: PyTorch, TensorFlow, FEniCS, Scikit-learn, SciPy, NumPy, Chebfun

Languages : English (Fluent), Mandarin (Native), Cantonese (Good)

Others : LaTeX, Microsoft Office, Git